(i) Simultaneous Firing

Consider the entire system (truck, guns, and n projectiles). Since there are no external horizontal forces, the total momentum $\vec{P}$ is conserved. Initially, the system is at rest, so $\vec{P}_{\text{initial}} = 0$.

After all n guns are fired simultaneously, let the speed of the truck be u. The speed of each projectile relative to the ground is v in the opposite direction (given the muzzle speed is $v - V$ and $V = 0$ initially). Thus:

$$\begin{aligned} 0 &= Mu - nmv \\ \implies u &= \frac{nmv}{M} \end{aligned}$$

The total kinetic energy K gained by the system is the sum of the kinetic energy of the n projectiles and the truck:

$$\begin{aligned} K &= \sum KE_{\text{projectiles}} + KE_{\text{truck}} \\ &= n\left(\frac{1}{2}mv^2\right) + \frac{1}{2}Mu^2 \\ &= \frac{1}{2}nmv^2 + \frac{1}{2}M\left(\frac{nmv}{M}\right)^2 \\ &= \frac{1}{2}nmv^2 + \frac{1}{2}\frac{n^2m^2v^2}{M} \\ &= \frac{1}{2}nmv^2\left(1 + \frac{nm}{M}\right) \quad \blacksquare \end{aligned}$$

(ii) Firing One at a Time

Let u_r be the speed of the truck after r guns have been fired. Consider the r-th firing. Immediately before this shot, the truck (with $n - r + 1$ projectiles) has speed u_{r-1}. The speed of the r-th projectile relative to the ground is $-(v - u_{r-1})$ (moving backwards). Immediately after, the truck (with $n - r$ projectiles) has speed u_r.

By conservation of momentum for this single event:

$$[M + (n - r + 1)m]u_{r-1} = -m(v - u_{r-1}) + [M + (n - r)m]u_r$$

Expanding and rearranging terms:

$$\begin{aligned} (M + (n - r)m)u_{r-1} + mu_{r-1} &= -mv + mu_{r-1} + (M + (n - r)m)u_r \\ (M + (n - r)m)u_r - (M + (n - r)m)u_{r-1} &= mv \\ u_r - u_{r-1} &= \frac{mv}{M + (n - r)m} \quad (*) \end{aligned}$$

To show $u_n < u$: Note that $u_n = \sum_{r=1}^{n}(u_r - u_{r-1})$. From $(*)$, for $1 \le r < n$, we have:

$$M + (n - r)m > M \implies \frac{mv}{M + (n - r)m} < \frac{mv}{M}$$

Summing from $r = 1$ to n:

$$u_n = \sum_{r=1}^{n} \frac{mv}{M + (n - r)m} < \sum_{r=1}^{n} \frac{mv}{M} = \frac{nmv}{M} = u$$

Thus, $u_n < u$. $\blacksquare$

(iii) Kinetic Energy Analysis

Let K_r be the total kinetic energy after r shots. The gain in energy during the r-th shot is:

$$\begin{aligned} K_r - K_{r-1} &= KE_{\text{projectile}} + \Delta KE_{\text{truck}} \\ &= \frac{1}{2}m(v-u_{r-1})^2 + \frac{1}{2}[M+(n-r)m]u_r^2 - \frac{1}{2}[M+(n-r+1)m]u_{r-1}^2 \end{aligned}$$

Substituting $u_r = u_{r-1} + \frac{mv}{M_r}$ (where $M_r = M + (n-r)m$):

$$\begin{aligned} K_r - K_{r-1} &= \frac{1}{2}m(v^2 - 2vu_{r-1} + u_{r-1}^2) + \frac{1}{2}M_r\left(u_{r-1} + \frac{mv}{M_r}\right)^2 - \frac{1}{2}(M_r+m)u_{r-1}^2 \\ &= \frac{1}{2}mv^2 - mvu_{r-1} + \frac{1}{2}mu_{r-1}^2 + \frac{1}{2}M_r\left(u_{r-1}^2 + \frac{2mu_{r-1}v}{M_r} + \frac{m^2v^2}{M_r^2}\right) - \frac{1}{2}M_ru_{r-1}^2 - \frac{1}{2}mu_{r-1}^2 \\ &= \frac{1}{2}mv^2 - mvu_{r-1} + mvu_{r-1} + \frac{1}{2}\frac{m^2v^2}{M_r} \\ &= \frac{1}{2}mv^2 + \frac{1}{2}mv\left(\frac{mv}{M_r}\right) \end{aligned}$$

Since $u_r - u_{r-1} = \frac{mv}{M_r}$, we obtain:

$$K_r - K_{r-1} = \frac{1}{2}mv^2 + \frac{1}{2}mv(u_r - u_{r-1})$$

Summing this expression from $r = 1$ to n:

$$\sum_{r=1}^{n}(K_r - K_{r-1}) = \sum_{r=1}^{n}\frac{1}{2}mv^2 + \sum_{r=1}^{n}\frac{1}{2}mv(u_r - u_{r-1})$$

$$K_n - K_0 = \frac{1}{2}nmv^2 + \frac{1}{2}mv(u_n - u_0)$$

Given $K_0 = 0$ and $u_0 = 0$:

$$K_n = \frac{1}{2}nmv^2 + \frac{1}{2}mvu_n \quad \blacksquare$$

Deduction of $K_n < K$:
From part (i), $K = \frac{1}{2}nmv^2 + \frac{1}{2}mv\left(\frac{nmv}{M}\right) = \frac{1}{2}nmv^2 + \frac{1}{2}mvu$.
Since we proved in part (ii) that $u_n < u$, it directly follows that:

$$\frac{1}{2}nmv^2 + \frac{1}{2}mvu_n < \frac{1}{2}nmv^2 + \frac{1}{2}mvu \implies K_n < K \quad \blacksquare$$

This solution was submitted by **Roy**

Published and refined by **Roy**

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